Uzair Arif

uarif2093@gmail.com | linkedin.com/in/uzairarif | uzairlol.github.io | github.com/uzairlol | +92 349 6323414

Final-year Economics & Mathematics student at IBA Karachi with hands-on experience building LLM systems, time-series forecasting pipelines, and production RAG applications. Interned at Pakistan's Federal Board of Revenue and NASTP. Looking for junior AI/ML roles.

Education

Institute of Business Administration — BS Economics and Mathematics, CGPA: 3.19 *June 2026*

Key Courses: Foundations of Data Science, Linear Algebra, Discrete Mathematics, Statistical Machine Learning

Skills

Languages & Libraries: Python, SQL, Scikit-learn, XGBoost, CatBoost, PyTorch, Prophet, Darts, ZeroMQ

LLM / RAG: LangChain, FAISS, BM25, Reciprocal Rank Fusion, Groq (Llama 3), SciBERT, Optuna

Tools & Platforms: QGIS, Streamlit, Jupyter, Git, Kaggle

Core Concepts: Time-Series Forecasting, Anomaly Detection, NLP, RAG, Multi-Agent Simulation

Professional Experience

Federal Board of Revenue (FBR) — Revenue Forecasting Intern *Jun 2025 – Jul 2025*

- Built 5 daily Prophet models (Income Tax, Sales Tax, Federal Excise, Customs Duty, Gross Collection) with Pakistan-specific Gregorian and lunar holiday effects and monthly seasonality.
- Tuned all models via Optuna hyperparameter search and evaluated volatile revenue patterns on held-out validation sets using MAE and MSE.
- Automated a multi-horizon Excel reporting pipeline with daily, weekly, and monthly breakdowns; deployed an interactive Streamlit dashboard to surface mid-month surges and fiscal deadline spikes for the team.

Ali Khan Consultants — Junior Data Analyst *Sep 2024 – Dec 2024*

- Identified 32% (60,000 kg) forecast discrepancy in procurement records, driving order delays and supply chain disruptions.
- Diagnosed 15-day gap from process to sales order (above industry avg) and 4-day lag to dispatch, highlighting warehousing inefficiencies.
- Found 41% of monthly revenue occurs in last 5 days; final-day orders 3.5× above expected daily average.

National Aerospace Science & Technology Park (NASTP) — AI Intern *Jun 2024 – Aug 2024*

- Engineered real-time data streaming and visualization tools using ZeroMQ, reducing publish-subscribe latency to milliseconds.
- Built CNNs, linear regression, and SVM models from scratch for aerospace sensor data classification.
- Created an anomaly detection model for ECG time-series data, achieving a 95% detection rate for anomalies.

Projects

Multi-Agent LLM Simulation for Climate Governance

- Simulated 26 heterogeneous LLM agents (Llama 3.1 8B) in a 15-round repeated public goods game modelling the Loss & Damage Fund; agents overcame individually irrational contribution incentives (MPCR ≈ 0.062) through reputation feedback and visible accounting, with cooperation rising 1,390× from round 1 to round 15.
- Found a 1.2% disbursal rate — the fund accumulated $34.5B in contributions but paid out only $424M — revealing that the LDF operates as a savings vehicle rather than redistributive insurance, with net transfer to developing nations of just $204M despite $34.4B in developed-group contributions.
- Designed deterministic institutional assignment (developed agents to Sanctioning Institution, developing agents to Sanction-Free Institution) reflecting real-world normative asymmetries; Gini declined modestly from 0.71 to 0.61 across rounds, showing cooperation does not automatically produce equitable redistribution under proportional payout rules.

Impostor Hunt: Real vs. Fake Text Classification

- Achieved public score of 0.90663 in ESA's Impostor Hunt competition (Secure Your AI series), detecting hallucinated and data-poisoned LLM outputs from faithful ones — trained on only 95 samples against 1068 test pairs.
- Recovered from mid-competition label correction that collapsed scores from ∼0.78 to ∼0.44; re-engineered features (SciBERT + PCA, GPT-2 perplexity, bigram Jaccard, KL divergence, burstiness) re-tuned via Optuna + 5-fold CV to reach final score.

IBA RAG Chatbot

- Built a modular, production-structured hybrid RAG system (9-module Python package) combining FAISS semantic search and BM25 keyword retrieval fused via Reciprocal Rank Fusion across 1000+ HTML files and PDF documents.
- Applied recursive chunking with 200-token overlap to prevent context loss across section boundaries.
- Integrated Groq LPU inference backend to achieve sub-second retrieval-augmented generation latency across documents.

Certifications

Generative AI: Working with LLMs | Introduction to LLMs | MLOps Essentials *NASBA*